

QUANTITATIVE ECONOMICS - SPRING 2022

Answer any FIVE questions.

1. Consider the regression model

$$Y_i = \beta X_i + u_i \quad i = 1, \dots, n$$

where u_i and X_i satisfy the standard assumptions. Let $\bar{\beta} = \frac{\bar{Y}}{\bar{X}}$, where $\bar{Y}$ and $\bar{X}$ are the sample means of Y_i and X_i , respectively.

- (a) Show that $\bar{\beta}$ is a linear function of $Y_1, Y_2, \dots, Y_n$.
- (b) Show that $\bar{\beta}$ is conditionally unbiased.
- (c) Derive the variance of $\bar{\beta}$.

2. Consider the regression model

$$y_i = \alpha + \beta x_i + u_i, \quad i = 1, 2, \dots, n$$

- (a) Suppose we assume that $\alpha = 0$ and hence $y_i = \beta x_i + u_i$
 - i. Write down the moment condition(s) used to derive the method of moments estimator for β .
 - ii. Write down the least-squares objective function to derive the OLS estimator for β .
 - iii. Use the equation from ii. to derive the estimator for β .
- (b) Suppose we assume that $\beta = 0$ and hence $y_i = \alpha + u_i$
 - i. Write down the moment condition(s) used to derive the method of moments estimator for α .
 - ii. Write down the least-squares objective function to derive the OLS estimator for α .
 - iii. Use the equation from ii. to derive the estimator for α .

3. Suppose you collect data from a survey on wages, education, experience, and gender. In addition, you ask for information about marijuana usage. The original question is: "On how many separate occasions last month did you smoke marijuana?"

- (a) Write an equation that would allow you to estimate the effects of marijuana usage on wage, while controlling for other factors. You should be able to make statements such as, "Smoking marijuana five more times per month is estimated to change wage by x%."
- (b) Write a model that would allow you to test whether drug usage has different effects on wages for men and women. How would you test that there are no differences in the effects of drug usage for men and women?
- (c) Suppose you think it is better to measure marijuana usage by putting people into one of four categories: nonuser, light user (1 to 5 times per month), moderate user (6 to 10 times per month), and heavy user (more than 10 times per month). Now, write a model that allows you to estimate the effects of marijuana usage on wage.
- (d) Using the model in part (c), explain in detail how to test the null hypothesis that marijuana usage has no effect on wage. Be very specific and include a careful listing of degrees of freedom.

4. Consider the following data generating process

$$Y_t = \phi_3 Y_{t-3} + \varepsilon_t$$

where ε_t is a white noise process. Assuming $0 < \phi_3 < 1$, answer the following:

- (a) Derive the expected value of this process.
- (b) Derive the variance of this process.
- (c) Derive the covariance of this process for $j = 1, 2, 3$.
- (d) Derive the autocorrelation function of this process for $j = 1, 2, 3$.

5. Let variable “kids” denote the number of children ever born to a woman and let variable “educ” denote years of education for the woman.
 - (a) Write a simple regression model relating fertility to years of education. What kind of control variables you are going to use in your model?
 - (b) What kind of factors may be contained in unobservables? Are these likely to be correlated with level of education?
 - (c) Will a simple regression analysis uncover the ceteris paribus effect of education on fertility? Explain.
 - (d) What kind of data and identification strategy would address the causal link between the two variables? Explain in detail.

6. In 1985, neither Florida nor Georgia had laws banning open alcohol containers in vehicle passenger compartments. By 1990, Florida had passed such a law, but Georgia had not. Suppose you can collect random samples of the driving-age population in both states, for 1985 and 1990. Let arrest be a binary variable equal to unity if a person was arrested for drunk driving during the year.
 - (a) Using a diff-in-diff identification strategy, write down a linear probability model (LPM) that allows you to test whether the open container law reduced the probability of being arrested for drunk driving. Which coefficient in your model measures the effect of the law?
 - (b) What is the interpretation of the coefficient of interest in your LPM model?
 - (c) How will the interpretation be different if you use a Logit model instead of the LPM?
 - (d) What is the Parallel Trend Assumption and how would you check for this assumption in your data?

7. What is the difference between Sharp and Fuzzy RD strategy? Write down equations showing that a Fuzzy RD model leads to a simple 2SLS estimation strategy. Explain the first stage equation, the reduced form equation, the instrument, and the instrumented variables in your model.

8. Suppose we have data for U.S. interest rate data over the period 1960:Q1 to 2008:Q1 and we are interested in estimating a quarterly model of spread between a long-term and a short-term interest rate. Specifically, the interest rate spread (s) can be formed as the difference between the interest rate on a 10-year U.S. government bonds ($r10$) and the rate on a three-month treasury bills ($Tbill$) as

$$s_t = r10_t - Tbill_t \quad (1)$$

Suppose the interest rate spread in the fourth quarter of 2007 is 0.75, the interest rate spread in the first quarter of 2008 is 1.51 and the interest rate spread in the second quarter of 2008 is 1.61. Use the EViews output below to answer the following questions:

Dependent Variable: S Method: Least Squares Date: 12/30/09 Time: 10:28 Sample (adjusted): 1960Q3 2008Q1 Included observations: 191 after adjustments Convergence achieved after 3 iterations				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.387620	0.285227	4.864963	0.0000
AR(1)	1.109152	0.070829	15.65953	0.0000
AR(2)	-0.245386	0.070792	-3.466283	0.0007
R-squared	0.805374	Mean dependent var	1.378220	
Adjusted R-squared	0.803304	S.D. dependent var	1.210806	
S.E. of regression	0.536998	Akaike info criterion	1.609936	
Sum squared resid	54.21291	Schwarz criterion	1.661019	
Log likelihood	-150.7489	F-statistic	388.9784	
Durbin-Watson stat	1.960866	Prob(F-statistic)	0.000000	
Inverted AR Roots	.80	.31		

- Write the estimated equation given in the Eviews output.
- Obtain the one-step ahead forecast.
- Obtain the one-step ahead forecast error.
- Obtain the two-step ahead forecast.
- Obtain the three-step ahead forecast.